A Tableau for TCTL

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Abstract. We present the first tableau-based decision procedure for the satisfiability problem of Timed Computation Tree Logic (TCTL). Our approach is an on-the-fly, symbolic proof search that explores only the relevant part of the state space and excludes spurious Zeno witnesses, so that for every satisfiable formula it yields a realisable timed automaton as a certificate of satisfiability. Our procedure is sound and terminating for the general TCTL, and complete for its non-punctual fragment, where it matches the optimal ExpTime bound. Our evaluation shows that it remains feasible in practice, even for formulas with a large number of nested temporal operators, with a consistently low, almost-flat cost for the kind of TCTL properties engineers write in practice.

Keywords: TCTL · Satisfiability · Real-Time Systems

1 Introduction

Establishing the correctness of safety-critical software is a central concern of formal verification, for which model checking is the established approach [6,11]. For real-time systems, where correctness depends not only on functional behaviour but also on timing, software is typically modelled as a timed automaton (TA) [3] and verified against requirements expressed in Timed Computation Tree Logic (TCTL) [2] using efficient tools such as Uppaal [7], as exemplified by [13,16,18].

In this workflow, however, the verification step only establishes conformance between a system and a specification; it says nothing about the quality of the specification itself. A TCTL property may be unsatisfiable, valid or refined by another property in the sense of [9,10]. Each of these defects is ultimately a question of satisfiability, and detecting them therefore requires a satisfiability procedure for TCTL.

The satisfiability problem for TCTL is known to be undecidable in general [2,4]. Finite satisfiability is nonetheless recursively enumerable, so enumerating candidate automata and model checking each is sound and eventually finds a model for every satisfiable formula; it simply never terminates on an unsatisfiable one. La Torre and Napoli [14,15] traced the source of undecidability to punctuality, showing that once singular intervals are forbidden, i.e., in the *non-punctual* fragment, finite satisfiability becomes decidable in ExpTime by reduction to emptiness of timed tree automata, at the cost of a global construction, exponential in the formula. Subsequent work traded completeness for

implementability: Penczek et al. [19] reduce satisfiability to the existence of a *bounded* parametric TA, instantiated with IMITATOR [5]; their method is sound but incomplete, does not handle nested modalities, and terminates by virtue of the imposed bound.

To this day, no search-based, goal-directed procedure for TCTL satisfiability has been proposed, and the only practical approach is bounded and partial.

To fill this gap, we present the first on-the-fly, symbolic and tableau-based decision procedure for satisfiability of TCTL. Unlike previous work, not only is our procedure sound, but also terminating for the general TCTL. On the non-punctual fragment the procedure is in addition complete, matching the optimal EXPTIME bound [11,14]. Completeness moreover extends beyond that fragment, to every formula of temporal nesting depth 1, punctuality included, thus deciding most of the typical properties engineers write in practice. Finally, our procedure is directly implementable, matches the optimal EXPTIME bound [15], and returns a witnessing TA whenever the input formula is satisfiable.

Unlike the classical tableau constructions for logics such as CTL or *discrete*-time quantitative CTL [1,11,12], our calculus must contend with two phenomena of the dense-time semantics. First, *zenoness*: a cycle may be traversed infinitely often within a bounded total delay, so a branch defers a deadline for ever without ever reaching it, along a run no system exhibits. Second, an eventuality under a AG is demanded at a continuum of instants, each carrying its own deadline. We settle the first with an acceptance condition on the tableau graph itself, and the second by covering the continuum symbolically.

We structure the paper as follows. Section 2 presents the background; Section 3 presents our procedure; Section 4 establishes termination, soundness, completeness, and complexity; Section 6 evaluates a prototype.

2 Background

A time bound I is a non-empty interval of $\mathbb{R}_{\geq 0}$ with integer endpoints of the form $[a,b], (a,b], [a,b), (a,b)$ with $a \leq b$, $a \in \mathbb{Z}_{\geq 0}$, $b \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$. Non-emptiness leaves $[a,a]$ as the only bound with $a = b$; we call it *punctual*.

Fix a finite set $\mathcal{P}$ of atomic propositions. We study TCTL in negation normal form over these time bounds, built according to the following grammar:

$$\varphi ::= \top \mid \bot \mid p \mid \neg p \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \mathrm{E}(\varphi \mathbf{U}_I \varphi) \mid \mathrm{E}(\varphi \mathbf{R}_I \varphi) \mid \mathrm{A}(\varphi \mathbf{U}_I \varphi) \mid \mathrm{A}(\varphi \mathbf{R}_I \varphi).$$

Other operators are expressible by the grammar: $\mathrm{EF}_I\varphi \equiv \mathrm{E}(\top \mathbf{U}_I \varphi)$, $\mathrm{AF}_I\varphi \equiv \mathrm{A}(\top \mathbf{U}_I \varphi)$, $\mathrm{EG}_I\varphi \equiv \mathrm{E}(\bot \mathbf{R}_I \varphi)$, and $\mathrm{AG}_I\varphi \equiv \mathrm{A}(\bot \mathbf{R}_I \varphi)$. Unlike CTL the logic has no next operator: over dense time no configuration has an immediate successor [2]. The symbols EX, AX used by the tableau are proof-theoretic markers internal to the construction, never object operators. Throughout, we consider a formula φ of size $n = |\mathsf{FL}(\varphi)|$, where $\mathsf{FL}(\varphi)$ is its Fischer–Ladner closure.

TCTL is interpreted over the timed transition system (TTS) induced by a TA [2,6]. A TA [7,8] over set of clocks $\mathcal{C}$ is a tuple $\mathcal{A} = \langle S, s_0, \mathsf{Inv}, E, \mathcal{P}, L \rangle$: locations S, initial s_0, a location invariant $\mathsf{Inv}(s)$ (that is, a downward-closed

constraint on clock valuations), discrete edges $E \subseteq S \times 2^{\mathcal{C}} \times S$ (each carrying the clocks it resets), and a labelling $L : S \to 2^{\mathcal{P}}$. An invariant bounds how long time may elapse in a location: without it a run could idle in a location forever. Invariant-free locations are allowed, so the model expresses idling as well. $\mathcal{A}$ is non-blocking (time can always diverge, hence no timelock).

Any $\mathcal{A}$ induces a TTS $\mathcal{T} = \langle C, c_0, \dot{\to}, \to \rangle$, whose configurations $C = \{(s, \nu) \in S \times \mathbb{R}^{\mathcal{C}}_{\geq 0} \mid \nu \models \mathsf{Inv}(s)\}$ pair a location with an admissible clock valuation, and whose initial configuration is $c_0 = (s_0, \mathbf{0})$. Two relations act on configurations: $\dot{\to} \subseteq C \times \mathbb{R}_{\geq 0} \times C$, the delay relation, which we write $(s, \nu) \xrightarrow{t} (s, \nu + t)$ for $t \in \mathbb{R}_{\geq 0}$ such that $\nu + t' \models \mathsf{Inv}(s)$ throughout $t' \in [0, t]$; and $\to \subseteq C \times C$, the discrete relation, which we write $(s, \nu) \to (s', \nu[R := 0])$ for each edge $(s, R, s') \in E$ with $\nu \models \mathsf{Inv}(s)$ and $\nu[R := 0] \models \mathsf{Inv}(s')$, where the *reset* $\nu[R := 0]$ zeroes every clock in $R \subseteq \mathcal{C}$ and leaves the others unchanged.

A run ρ from any $c \in C$ is a maximal alternation of delays $t_0, t_1, \ldots$ and discrete steps; a position $\pi = (i, \delta)$ with $0 \leq \delta \leq t_i$ denotes the configuration $\rho(\pi)$ after the i-th discrete step plus delay δ, at elapsed time $\mathit{time}(\pi) = T_i + \delta$ (with $T_i = \sum_{j<i} t_j$), positions ordered by the lexicographic $\preceq$. The run is non-Zeno when its delays sum to ∞; by non-blockingness the set $\mathrm{Runs}^{\infty}(c)$ of non-Zeno runs is non-empty. The TCTL satisfaction relation is the standard one [2], evaluated over Runs^{∞}; for a single run, timed until and release read:

$$\rho \models \varphi \mathbf{U}_I \psi \iff \exists \pi \big[\mathit{time}(\pi) \in I \wedge \rho(\pi) \models \psi \wedge \forall \pi' \prec \pi (\rho(\pi') \models \varphi) \big] \tag{1}$$

$$\rho \models \varphi \mathbf{R}_I \psi \iff \forall \pi \big[\mathit{time}(\pi) \in I \Rightarrow \big(\rho(\pi) \models \psi \vee \exists \pi' \prec \pi (\rho(\pi') \models \varphi) \big) \big] \tag{2}$$

The path quantifier then lifts a run formula φ to a configuration: $(s, \nu) \models \mathrm{E}(\varphi)$ iff some $\rho \in \mathrm{Runs}^{\infty}(s, \nu)$ satisfies φ, and $(s, \nu) \models \mathrm{A}(\varphi)$ iff all do. Quantifying over positions rather than absolute time keeps the clauses unambiguous at discrete jumps; restricting to Runs^{∞} is the usual dense-time reading.

The formula φ is satisfiable, or SAT, if there exists a TA whose induced TTS satisfies it at its initial configuration, $\mathcal{T}, (s_0, \mathbf{0}) \models \varphi$, otherwise we say it is unsatisfiable, or UNSAT. We call such an automaton a witness for φ.

The tableau reasons symbolically about timed valuations. A zone is a set $Z \subseteq \mathbb{R}^{\mathcal{C}}_{\geq 0}$ defined by a finite conjunction of difference constraints $x - y \prec c$ (with $x, y \in \bar{\mathcal{C}} \cup \{0\}$, 0 a fictitious clock fixed at 0, and the bound $(c, \prec)$ ranging over $(\mathbb{Z} \times \{<, \leq\}) \cup \{(\infty, <)\}$), stored as a difference-bound matrix (DBM) [3,8]; the strictness flag is needed to represent the open time bounds the rules intersect with. Zones are closed under intersection, intersection with a single-clock guard $x \in I$, reset $Z[R := 0]$, and time-successor $\uparrow Z = \{\nu + t \mid \nu \in Z, t \geq 0\}$. Zones are compared modulo max-constant extrapolation: for M the largest constant in the input, α_M relaxes every bound above M, and we store each zone as the canonical (shortest-path-closed) DBM of $\alpha_M(Z)$, writing $Z \cong_M Z'$ when these canonical forms coincide, so $\cong_M$-equivalent zones are syntactically identical. We keep the tableau construction diagonal-free, so that max-constant extrapolation is sound [8].

3 The Tableau

We decide satisfiability of φ by an on-the-fly, symbolic tableau construction. We fix nodes and realisability (§3.1), give the rules (§3.2), define acceptance (§3.3), and close with two examples (§3.4).

3.1 Nodes and Realisability

A node is a tuple $N = \langle \Gamma, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle$, where Γ is the set of goal formulas to be satisfied (write $\Gamma_{\mathsf{prop}} = \Gamma \cap \big(\mathcal{P} \cup \{\neg p \mid p \in \mathcal{P}\}\big)$ for the propositional part); $\mathcal{X}$ the context of universal formulas that must hold at every successor; Z the set of clock valuations for this node, represented as a canonical zone; $\mathcal{O}$ the clock map, which allocates to every active timed modality μ a dedicated formula-clock x_μ so its window (i.e. its interval I) can be measured from the configuration that demanded it; and $\mathcal{D}$ is a set of deadlines $(x_\mu, \sup I)$, each with a strictness flag, and $\mathsf{Inv}(N) = \bigwedge_{(x_\mu, b) \in \mathcal{D}} x_\mu \leq b$ its induced invariant. Besides these five components a node records the clocks reset by the rule that produced it and, if that rule was a stop branch, the modality it discharged; a rule sets both from the branch it takes, and we leave them implicit in the displays below.

A node is *closed* if its zone is empty, $Z = \emptyset$, since it then has no valuations to realise, or if $\bot \in \Gamma_{\mathsf{prop}}$. It is *saturated* when no saturation or cover rule applies; only saturated nodes generate successors. It is *realisable* if there is a non-blocking TTS $\mathcal{T}$ induced by a TA, a location s with $L(s) \supseteq \Gamma_{\mathsf{prop}}$ and $\mathsf{Inv}(s) = \mathsf{Inv}(N)$, and an interpretation of the formula-clocks in $\mathcal{O}$, such that for every $\nu \in Z$ the configuration (s, ν) $\mathcal{T}, (s, \nu) \models \bigwedge \Gamma \ \wedge \ \bigwedge \mathcal{X}$. Realisability is monotone under narrowing: if $\langle \Gamma, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle$ is realisable and $\emptyset \neq Z' \subseteq Z$, then so is $\langle \Gamma, \mathcal{X}, Z', \mathcal{O}, \mathcal{D} \rangle$.

Two nodes N_1 and N_2 are *equivalent*, written $N_1 \equiv N_2$, when they agree in Γ, $\mathcal{X}$, $\mathcal{O}$, $\mathcal{D}$, in the clocks reset by the rule that produced them and in the modality that rule discharged, and their zones agree modulo $\cong_M$.

An *OR* (disjunctive, $\oplus$) node is realisable if *some* child is; an *AND* (conjunctive, $\otimes$) node requires *all* its children. Some rules are deterministic ($\odot$): a single child, replacing the node. We write $\mathit{kind}(N) \in \{\odot, \oplus, \otimes\}$.

3.2 Rules

Rules come in four groups: *saturation* decomposes goals, *cover* discharges and launches a timed modality, *pursuit* unfolds a launched one step by step, and *successor generation* expands a saturated node.

Saturation. Conjunction ($\wedge$) is deterministic, disjunction ($\vee$) branches, and (AX) handles a universal marker by moving its argument into the context.

$$\frac{\langle \Gamma \cup \{\varphi \wedge \psi\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle}{\langle \Gamma \cup \{\varphi, \psi\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle}\ {\scriptstyle (\wedge)\ \odot} \qquad \frac{\langle \Gamma \cup \{AX\varphi\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle}{\langle \Gamma, \mathcal{X} \cup \{\varphi\}, Z, \mathcal{O}, \mathcal{D} \rangle}\ {\scriptstyle (\mathrm{AX})\ \odot}$$

$$\frac{\langle \Gamma \cup \{\varphi \vee \psi\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle}{\begin{array}{c}\langle \Gamma \cup \{\varphi\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle \\ \langle \Gamma \cup \{\psi\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle\end{array}}\ {\scriptstyle (\vee)\ \oplus}$$

Pursuing time modalities. When Γ contains a timed modality $\mu = \mathrm{Q}(\varphi \circ_I \psi)$, with $\mathrm{Q} \in \{\mathrm{E}, \mathrm{A}\}$ and $\circ \in \{\mathbf{U}, \mathbf{R}\}$, we say the node *demands* μ, and μ must hold at every valuation of the node's zone. To fulfil the demand the rule produces a (discharge) child, which drops μ from Γ and lets the node continue with its remaining goals, and a (launch) child, which starts a *pursuit* of μ, written $\mu^\bullet$, i.e., a branch that unfolds μ step by step in search of a witness. The launch introduces a fresh formula-clock x_μ and resets it here, so that I is measured from this configuration.

$$\frac{\langle \Gamma \cup \{\mu\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\begin{array}{l}\langle \Gamma, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle \quad \textsf{(discharge)} \\ \langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, Z[x_\mu := 0], \mathcal{O}[\mu \mapsto x_\mu], \mathcal{D} \cup \mathcal{D}_\mu\rangle \ \textsf{(launch)}\end{array}} \begin{array}{l}\mathcal{O}(\mu) \text{ undefined, or} \\ \sup I < \infty \text{ and } \mu \text{ not} \\ \text{a universal until}\end{array} \begin{array}{c}\text{(Cov)} \\ \otimes\end{array}$$

A deadline $\mathcal{D}_\mu = \{(x_\mu, \sup I)\}$ is applied iff μ is a universal until, with $\mathcal{D}_\mu = \emptyset$ otherwise. Only such a modality needs one, as it obliges *every* run to satisfy ψ within I and time must be forced forward or a run could idle past the window with the eventuality unmet. An existential until is discharged by a single witnessing run, and a release, of either quantifier, is discharged by lapsing past the window rather than by acting within it.

Pursuing an Until. Consider an until modality $\mu = \mathrm{Q}(\varphi \mathbf{U}_I \psi)$, writing $a = \inf I$ and $b = \sup I$. An ordinary pursuit $\mu^\bullet$ unfolds by ($\mathbf{U}^\bullet$): either ψ holds now, inside the window, or φ holds now and the pursuit passes to the successors.

$$\frac{\langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\langle \Gamma \cup \{\psi\}, \mathcal{X}, Z \cap (x_\mu \in I), \mathcal{O}, \mathcal{D} \setminus \{(x_\mu, b)\}\rangle \quad (\mathbf{U}\text{-stop})} \mu = \mathrm{Q}(\varphi \mathbf{U}_I \psi) \ \begin{array}{c}(\mathbf{U}^\bullet) \\ \oplus\end{array}$$

$$\begin{cases} \langle \Gamma \cup \{\varphi, EX\mu^\bullet\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle & \text{if } \mathrm{Q} = \mathrm{E}, \\ \langle \Gamma \cup \{\varphi\}, \mathcal{X} \cup \{AX\mu^\bullet\}, Z, \mathcal{O}, \mathcal{D}\rangle & \text{if } \mathrm{Q} = \mathrm{A}, \end{cases} \quad (\mathbf{U}\text{-go})$$

($\mathbf{U}$-go) defers the pursuit to the successors, where x_μ keeps running and the window stays open; for $\mathrm{Q} = \mathrm{E}$ a single witnessing run suffices, so the residual travels as a goal to one successor, and for $\mathrm{Q} = \mathrm{A}$ every run must comply, so it travels as context to all of them. ($\mathbf{U}$-stop) ends the pursuit: $\mu^\bullet$ is dropped from Γ and its deadline removed from $\mathcal{D}$, the single onset it was launched for having been answered.

Pursuit must end when the window closes, i.e., $x_\mu \in I$ fails. The rule ($\mathbf{U}$-miss) empties the zone and closes the node, which is then no longer realisable.

$$\frac{\langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, \emptyset, \mathcal{O}, \mathcal{D}\rangle} \begin{array}{l} x_\mu \notin I, \\ b < \infty \end{array} \begin{array}{c}(\mathbf{U}\text{-miss}) \\ \odot\end{array}$$

For a universal until the deadline in $\mathcal{D}$ empties the zone first and the rule never fires.

As shown in Fig. 1, a demand can be potentially requested by a continuum of onsets. We handle it symbolically, according to whether $\sup I < \infty$ or not. For $\sup I = \infty$, the new onset is absorbed into the running one: (Abs) drops μ from Γ, so it is not demanded again, and keeps the previous pursuit alive. A new

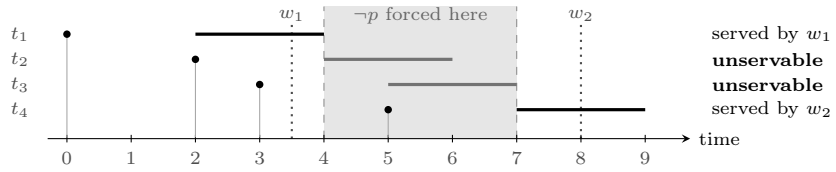


Fig. 1: The formula $\mathrm{AG}\,\mathrm{EF}_{[2,4]}\,p \wedge \mathrm{AG}_{[4,7]}\neg p$ (UNSAT). The bar at t_i is that onset's window $[t_i+2, t_i+4]$, and a p at w serves the onsets whose bar contains it. The pursuit launched at t_1 stops at w_1 and serves no later onset.

one is launched only when the previous pursuit stops, so a run carries a stream of witnesses, and every onset here asks only for a ψ at some later point, which the stream supplies.

For $\sup I < \infty$, some onsets can not be absorbed, as shown in Fig. 1. In this case, we distinguish between the existential until, which is simply pursued by a new (Cov), and a universal until, which is *carried*. Every run must comply with the demand, so one pursuit that witnesses at least every $b-a$ covers every onset and it turns *persistent*, written μ^*, which is unfolded by ($\mathbf{U}^*$).

$$\frac{\langle \Gamma \cup \{\mu, \mu^\bullet\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\langle \Gamma \cup \{\mu^*\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}\ \begin{array}{l}\mu \text{ universal} \\ \text{until, (Carry)} \\ b<\infty \quad \odot\end{array} \qquad \frac{\langle \Gamma \cup \{\mu, \mu^\dagger\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\langle \Gamma \cup \{\mu^\dagger\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}\ \begin{array}{l}\mu^\dagger \in \{\mu^\bullet, \mu^*\}, \\ b=\infty \text{ or (Abs)} \\ \mu^\dagger = \mu^* \quad \odot\end{array}$$

$$\frac{\langle \Gamma \cup \{\mu^*\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\begin{array}{l}\langle \Gamma \cup \{\psi, \mu^*\}, \mathcal{X}, Z \cap (x_\mu \in I^*)[x_\mu := 0], \mathcal{O}, \mathcal{D}[x_\mu \mapsto b-a]\rangle\ (\mathbf{U}^*\text{-stop}) \\ \langle \Gamma \cup \{\varphi\}, \mathcal{X} \cup \{AX\mu^*\}, Z, \mathcal{O}, \mathcal{D}\rangle\ (\mathbf{U}^*\text{-go})\end{array}}\ \begin{array}{l}(\mathbf{U}^*) \\ b<\infty \quad \oplus \\ \mu = \mathrm{A}(\varphi \mathbf{U}_I \psi)\end{array}$$

($\mathbf{U}^*$-stop) keeps the pursuit alive. A persistent pursuit answers every onset carried into it, and the context keeps raising more, so μ^* stays and its clock restarts: x_μ is re-zeroed, measuring from this occurrence of ψ, and its deadline is set to $b-a$. That bound is the length of the window. A time w at which ψ holds answers exactly the onsets in $[w-b, w-a]$, so two consecutive such times leave no onset unanswered precisely when they lie within $b-a$ of each other. The guard follows the same reading: I^* is I at the first witness, where x_μ still measures from the onset that launched the pursuit, and $[0, b-a]$ at every later one, where it measures from the preceding witness.

Pursuing a Release. Consider now a release modality $\mu = \mathrm{Q}(\varphi \mathbf{R}_I \psi)$. The semantics of release imposes that ψ must hold at every point in the window, unless φ releases it. We handle the pursuit of a release in two stages. First, we partition the zone according to the position of x_μ relative to I, and then we pursue each part separately. We write $Z^{\prec I} = Z \cap (x_\mu < \inf I \text{ or } x_\mu = \inf I \notin I)$, $Z^{\in I} = Z \cap (x_\mu \in I)$ and $Z^{\succ I} = Z \cap (x_\mu > \sup I \text{ or } x_\mu = \sup I \notin I)$ for the three parts of the partition, so that the strictness of the interval endpoints is carried by the DBM bounds and the three parts are a partition of Z. Empty parts are dropped. Accordingly, the unfolding of a release produces up to three children,

one for each part of the partition. The first two are pursued, while the last is lapsed and drops $\mu^\bullet$.

$$\frac{\langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\begin{array}{ll}\langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, Z^{\prec I}, \mathcal{O}, \mathcal{D}\rangle & (\mathbf{R}^\bullet_{\prec I})\\ \langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, Z^{\in I}, \mathcal{O}, \mathcal{D}\rangle & (\mathbf{R}^\bullet_{\in I})\\ \langle \Gamma, \mathcal{X}, Z^{\succ I}, \mathcal{O}, \mathcal{D}\rangle & (\mathbf{R}^\bullet_{\succ I})\end{array}} \quad \mu = \mathrm{Q}(\varphi \mathbf{R}_I \psi) \overset{(\mathbf{R}^{\mathsf{split}})}{\otimes}$$

The $(\mathbf{R}^\bullet_{\prec I})$ and $(\mathbf{R}^\bullet_{\in I})$ parts are pursued by the following rule, which, similarly to $(\mathbf{U}^\bullet)$ is a disjunction between two cases: either φ is released, or the pursuit continues.

$$\frac{\langle \Gamma \cup \{\mu^\bullet\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\langle \Gamma \cup \Theta \cup \{\varphi\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle \ (\mathbf{R}^\bullet\text{-stop})} \quad Z \subseteq Z^{\prec I} \text{ and } \Theta = \emptyset, \text{ or } Z \subseteq Z^{\in I} \text{ and } \Theta = \{\psi\} \overset{(\mathbf{R}^\bullet)}{\oplus}$$

$$\begin{cases} \langle \Gamma \cup \Theta \cup \{EX\mu^\bullet\}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle \text{ if } \mathrm{Q} = \mathrm{E}, \\ \langle \Gamma \cup \Theta, \mathcal{X} \cup \{AX\mu^\bullet\}, Z, \mathcal{O}, \mathcal{D}\rangle \text{ if } \mathrm{Q} = \mathrm{A}, \end{cases} (\mathbf{R}^\bullet\text{-go})$$

Successor generation. A saturated and covered node is expanded by two successor rules.(Delay) advances the zone to the next integer hyperplane $x = b$ ($b \leq M$) of an active bound, and does so in two steps: $\Delta^\circ Z$ is the time-successor of Z strictly below the next hyperplane, and $\Delta^\partial Z$ is that hyperplane (equivalently $\uparrow Z$ when no hyperplane remains at or below M). The hyperplanes of x_μ are the endpoints of I together with their *shifts* $\inf I - \inf J$ and $\sup I - \sup J$ by the window J of every other modality of φ. The shifts are what keep onsets apart: a μ banning ψ over I leaves a demand for ψ over J unsatisfiable exactly at the onsets in $[\inf I - \inf J, \sup I - \sup J]$, and splitting x_μ at the endpoints of I alone leaves that interval inside a cell that also holds satisfiable onsets, whose witness then discharges the whole cell. There are two shifts per pair of modalities and none depends on the magnitude of a constant, so the count of hyperplanes stays quadratic in $|\varphi|$ and flat in M.

$$\frac{\langle \Gamma, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\begin{array}{ll}\langle \Gamma, \mathcal{X}, \Delta^\circ Z \cap \mathsf{Inv}(N), \mathcal{O}, \mathcal{D}\rangle & (\mathsf{openseg.})\\ \langle \Gamma, \mathcal{X}, \Delta^\partial Z \cap \mathsf{Inv}(N), \mathcal{O}, \mathcal{D}\rangle & (\mathsf{boundary})\end{array}} \quad \begin{array}{l}\Gamma \text{ saturated}\\ \text{and covered}\end{array} \overset{(\text{Delay})}{\otimes}$$

The two children carry the same demands and differ only in their zone, and that difference is what a guard is tested against. Take a pursuit of $\mathrm{E}(p\mathbf{U}_{[1,2)}q)$ on the zone $x_\mu \in (1,2)$, with $x_\mu = 2$ the next hyperplane. On (openseg.) the ($\mathbf{U}$-stop) guard $x_\mu \in [1,2)$ is non-empty, so q may still be witnessed; on (boundary) it is empty, the window having closed, and only ($\mathbf{U}$-go) survives with an eventuality that can no longer be met. A single zone spanning both would have to answer once for two regions, and $[1,2)$ would become indistinguishable from $[1,2]$. Splitting keeps them apart at no cost in the extracted automaton, where a chain of delay steps is one continuous elapse within a single location.

(EX) takes the discrete steps and discharges the existential markers, one child per demanded $EX\psi$, each carrying ψ under the full context:

$$\frac{\langle \Gamma, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle}{\big(\ \langle \{\psi\} \cup \mathcal{X}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D}\rangle\ \big)_{EX\psi \in \Gamma}} \quad \begin{array}{l}\Gamma \text{ saturated}\\ \text{and covered}\end{array} \overset{(\text{EX})}{\otimes}$$

When no $EX\psi \in \Gamma$, a single *stutter* child $\langle \mathcal{X}, \mathcal{X}, Z, \mathcal{O}, \mathcal{D} \rangle$ keeps the run alive. Each child inherits $\mathcal{X}$, allowing (AX) to take effect, and it is why universal obligations constrain the whole branching rather than one path. A pursuit's residual rides both kinds of successor: $EX\mu^\bullet$ in Γ reaches one, $AX\mu^\bullet$ in $\mathcal{X}$ reaches all. So x_μ advances by elapsing time as well as across edges. A timed window is therefore entered by delay, its in-window stop taken at the first successor that satisfies it; a pure-delay witness, i.e., waiting in one location until $x_\mu \in I$, then stopping, is a chain of time-successors ending in a stop, the stutter step supplying a discrete anchor only when the location must be left.

3.3 The Tableau Graph and Acceptance

Through application of the rules, the tableau builds a graph of nodes, which can be used to establish satisfiability.

To build this graph we settle which rule fires at a node, since rule application is non-deterministic: several rules may apply to a node, which inherits the kind of the rule fired at it, and one rule may apply to several goals. Writing $\mathsf{T}(\varphi)$ for the timed modalities of $\mathsf{FL}(\varphi)$ and $P = \mathsf{FL}(\varphi) \cup \{\mu^\bullet, \mu^* \mid \mu \in \mathsf{T}(\varphi)\}$, we define $\mathsf{FL}^\bullet(\varphi) = P \cup \{EX\chi, AX\chi \mid \chi \in P\}$, with $|\mathsf{FL}^\bullet(\varphi)| = O(|\mathsf{FL}(\varphi)|)$. No rule produces a goal or a context formula outside $\mathsf{FL}^\bullet(\varphi)$.

We fix, arbitrarily, a linear order $\prec$ on formulas (e.g., lexicographic) and fire on the $\prec$-least goal of Γ to which a rule applies: a conjunction ($\wedge$), a disjunction ($\vee$), an $AX\varphi$ (AX), a demanded modality (Cov), (Carry) or (Abs) as their side conditions prescribe, an until pursuit past a bounded window (**U**-miss), an ordinary until pursuit $\mu^\bullet$ ($\mathbf{U}^\bullet$) and a persistent one μ^* ($\mathbf{U}^*$), and a release pursuit ($\mathbf{R}^\bullet$) when its zone lies wholly before or wholly inside the window and ($\mathbf{R}^{\mathsf{split}}$) otherwise.

This strategy produces for each node N a set of children $ch(N)$. A closed node has no children, and the children of a saturated one are the (Delay)- and the (EX)-children together, both successor rules being conjunctive, so it is an $\otimes$-node.

The tableau graph $\mathcal{G}_\varphi = (\mathcal{N}_\varphi, \mathcal{E}_\varphi)$ is then constructed from the nodes reachable under ch from the root $N_0 = \langle \{\varphi\}, \emptyset, \{\mathbf{0}\}, \emptyset, \emptyset \rangle$: its vertices V are those nodes up to $\equiv$, and it has an edge $e \in E$ from N to every $N' \in ch(N)$. A *path* of $\mathcal{G}_\varphi$ is a sequence of vertices, each a child of the preceding one, and a *cycle* is a path returning to its first vertex.

Lemma 1. *$|\mathcal{N}_\varphi| \leq 2^{O(n)} \cdot M^{O(n^2)}$, and every node has $O(n)$ children.*

Proof. Each component of a node ranges over a set of size $2^{O(n)}$: Γ and $\mathcal{X}$ over subsets of $\mathsf{FL}^\bullet(\varphi)$; $\mathcal{O}$ over subsets of $\mathsf{T}(\varphi)$, being determined by its domain; $\mathcal{D}$ over the assignments to each x_μ of $\sup I$ or $\sup I - \inf I$ with a strictness flag; the reset set over subsets of $\mathcal{C}$; and the discharged modality over $\mathsf{T}(\varphi)$. A $\cong_M$-zone over $K \leq n+1$ clocks is a $(K+1)^2$ DBM whose entries are ∞ or lie in $\{-M, \ldots, M\} \times \{<, \leq\}$, so there are $M^{O(n^2)}$ of them. Thus the bound is $2^{O(n)} \cdot M^{O(n^2)}$. Every rule has at most three children except (EX), with one per $EX\psi \in \Gamma$ and $|\Gamma| = O(n)$,.

A *marking* $\mathcal{M} \subseteq \mathcal{N}_\varphi$ is a set of vertices containing no closed node and containing, of each of its members, at least one child if that node is an $\oplus$-node and all of them if it is an $\otimes$-node. A cycle of $\mathcal{M}$ is *time-progressing* if it elapses positive time and every clock is either reset on it or extrapolated above M.

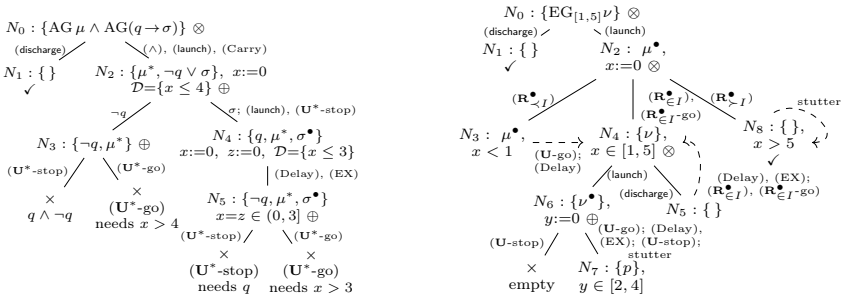
$\mathcal{M}$ is *successful* if every $N \in \mathcal{M}$ satisfies **(M1)** some path of $\mathcal{M}$ from N meets a time-progressing cycle; and **(M2)** for every until pursuit μ active at N, some path of $\mathcal{M}$ from N that carries it throughout, with x_μ unreset along the way, takes a stop branch ((**U**-stop) or (**U***-stop)) at a node N_k with $Z_k \cap (x_\mu \in I) \neq \emptyset$. A tableau *succeeds* on φ if N_0 lies in a successful marking.

(M2) ensures that an until pursuit is fulfilled rather than deferred for ever through (**U**-go). Its clause on x_μ ties the stop to the onset the pursuit measures: a relaunch on the way restarts the clock, and a stop taken past it witnesses the later onset. **(M1)** excludes Zeno witnesses. A clock never reset on a cycle and bounded there by some $k \leq M$ caps the delays of all its traversals at k, so infinitely many traversals elapse only finitely much time. The run they describe lies outside Runs^∞, on which satisfaction is defined, and is a behaviour no system exhibits. The requirement that a cycle elapse positive time rules out cycles built from expansion steps alone. Consider a persistent pursuit that has already carried over once: ψ is a goal, $x_\mu = 0$, and the deadline on x_μ is $b - a$. If $0 \in I$, a second application of (**U***-stop) yields the same node, since the goal it adds is already present, the clock it resets is already zero, and the deadline it sets is already in force. The node is its own child, and the resulting self-loop elapses no time, so **(M1)** keeps the node only if another path leaves it.

3.4 Two Worked Runs of The Tableau

Consider the formula $\mathrm{AG}\,\mu \wedge \mathrm{AG}(q \to \sigma)$ with $\mu = \mathrm{A}(\top \mathbf{U}_{[1,4]} q)$, that is $\mathrm{AF}_{[1,4]} q$, and $\sigma = \mathrm{A}(\bot \mathbf{R}_{(0,3]} \neg q)$, that is $\mathrm{AG}_{(0,3]}(\neg q)$. The first conjunct demands a q within $[1, 4]$ of every configuration; the second forbids a q for 3 time units after each one. The left conjunct proceeds as a persistent demand. Its first demand launches $\mu^\bullet$ with x reset and the deadline $x \leq \sup I = 4$; the next re-demand meets a live pursuit, so (Carry) folds it in and the pursuit becomes μ^*. Meanwhile the right conjunct re-imposes its body $\neg q \vee \sigma$ at every configuration, so N_2 carries both, and the disjunction is settled before μ^* is unfolded. Take $\neg q$, at N_3. The pursuit can then neither take (**U***-stop) nor (**U***-go): (**U***-stop) would assert q into a goal set already holding $\neg q$, and (**U***-go) can only defer, which past $x = 4$ empties the zone against the deadline. Both children close. Take σ instead. It is launched with its own clock z reset, and (**U***-stop) asserts q, re-zeroing x and recapping it at $b - a = 3$; this is N_4, where the two clocks begin to run together, $x = z$. That is the contradiction. The carried deadline $x \leq 3$ says the next q is due within 3, while σ forbids q for as long as $z \in (0, 3]$. At N_5 the (**U***-stop) branch needs a q it cannot have, and (**U***-go) needs $x > 3$, which the deadline denies. Both children close, so N_2 closes and the formula is UNSAT.

Consider $\mathrm{EG}_{[1,5]}\,\nu$, where $\nu = \mathrm{E}(\top \mathbf{U}_{[2,4]} p)$. Write x for the clock of the release, and y for that of ν. (Cov) splits N_0 into a discharge child, which carries no pursuit, and the launch N_2, where x is reset. ($\mathbf{R}^{\mathsf{split}}$) then partitions the zone,



(a) $\mathrm{AG}(\mathrm{AF}_{[1,4]}q) \wedge \mathrm{AG}(q \to \mathrm{AG}_{(0,3]}(\neg q))$. (b) $\mathrm{EG}_{[1,5]}\mathrm{E}(\top \mathbf{U}_{[2,4]}p)$.

Fig. 2: Two tableau runs. ✓ marks an open node, × a closed one.

conjunctively, into its three parts. At N_3, the ($\mathbf{R}^\bullet_{\prec I}$) part, (Delay) carries x into I. N_8, where $x > 5$, drops $\mu^\bullet$ and leaves a goal-free node that may delay indefinitely: a time-progressing loop, since no until needs to be fulfilled and x is extrapolated above M. N_4 represents the ($\mathbf{R}^\bullet_{\in I}$) part of ($\mathbf{R}^{\mathsf{split}}$), where ν is inserted into Γ and so demanded. (Cov) applies, giving two conjunctive branches. The (launch) child N_6 proves ν. It carries $\nu^\bullet$ with y reset; its ($\mathbf{U}$-stop) is empty there, the window $[2, 4]$ not yet open, so only ($\mathbf{U}$-go) survives until a delay carries y into $[2, 4]$, where ($\mathbf{U}$-stop) asserts p and drops $\nu^\bullet$. The (discharge) child N_5 drops ν from Γ and carries $\mu^\bullet$ on; a delay and (EX) revive the release, still inside $[1, 5]$, and its in-part demands ν once more. The search is back at N_4, closing a loop. That loop is not time-progressing: x is never reset, and the in-window zone holds it below 5. Nonetheless, we can construct a successful marking as $\mathcal{M} = \{N_0, N_1, N_2, N_3, N_4, N_5, N_6, N_7, N_8\}$. **(M2)** holds because the only until pursuit is $\nu^\bullet$, active at N_6, and the path to N_7 takes its ($\mathbf{U}$-stop) at a node whose zone meets $y \in [2, 4]$. **(M1)** holds because every node of $\mathcal{M}$ reaches a time-progressing loop: the nodes carrying the release reach N_8 by delaying until x passes 5, which lapses it, while N_1 and, once $\nu^\bullet$ is dropped, N_7 carry no pursuit, so no deadline caps their delays and their stutter loops are time-progressing. So $\mathcal{M}$, as well as the tableau, are successful and $\mathrm{EG}_{[1,5]}\,\nu$ is SAT.

4 The Decision Procedure

Section 3 reduces satisfiability of φ to whether N_0 lies in a successful marking of $\mathcal{G}_\varphi$. The procedure that decides this rests on reading the conditions of §3.3 as a removal operator F on subsets of $\mathcal{N}_\varphi$. For $A \subseteq \mathcal{N}_\varphi$, a node $v \in A$ survives when

- **(D1)** v is not closed
- **(D2)** $ch(v) \cap A \neq \emptyset$ if $kind(v) = \oplus$, and $ch(v) \subseteq A$ if $kind(v) = \otimes$
- **(D3)** v satisfies **(M1)** and **(M2)** with its paths taken inside A

that is, $F(A) = \{v \in A \mid v$ satisfies **(D1)**–**(D3)** on $A\}$.

Algorithm 1: Satisfiability check for TCTL formula φ

Input: φ in negation normal form
Output: SAT or UNSAT

```
1  E ← {N0};    Acc ← ∅;    Rej ← ∅
2  repeat
3  |  Acc ← Acc ∪ {v ∈ E | kind(v) = ⊕, ch(v) ∩ Acc ≠ ∅}
4  |  Rej ← Rej ∪ {v ∈ E | kind(v) = ⊗, ch(v) ∩ Rej ≠ ∅}
5  |  foreach component C of E \ (Acc ∪ Rej) with ch(C) \ C ⊆ Acc ∪ Rej do
6  |  |  X ← C
7  |  |  repeat X ← F(X ∪ Acc) ∩ C until X is unchanged
8  |  |  Acc ← Acc ∪ X;    Rej ← Rej ∪ (C \ X)
9  |  E ← E ∪ ch(E \ (Acc ∪ Rej))              // apply the rules of §3.2
10 until N0 ∈ Acc ∪ Rej
11 return SAT if N0 ∈ Acc, else UNSAT
```

F is monotone by construction. Let $A \subseteq B \subseteq \mathcal{N}_\varphi$ and let $v \in F(A)$. Then $v \in B$, and each of **(D1)**–**(D3)** still holds of v on B: **(D1)** does not mention the set; a $\oplus$-node keeps in B the child it had in A, and an $\otimes$-node has all its children in $A \subseteq B$; and a path inside A is a path inside B, which settles **(D3)** and **(D3)**. Hence $v \in F(B)$ and $F(A) \subseteq F(B)$.

Lemma 2. *$\mathcal{G}_\varphi$ has a greatest successful marking $\mathcal{M}^*$ and succeeds iff $N_0 \in \mathcal{M}^*$.*

Proof. Since F is monotone and $\mathcal{G}_\varphi$ is finite by Lemma 1, by Knaster–Tarski, F has a greatest fixpoint νF. Every marking surviving **(D1)**–**(D3)** is successful by construction, hence νF is the greatest successful marking. As a consequence, the tableau succeeds on φ iff $N_0 \in \nu F$.

A procedure that naively iterates F downwards from $\mathcal{G}_\varphi$ reaches νF in finitely many steps, but presupposes the full construction of $\mathcal{G}_\varphi$. Instead, Algorithm 1 decides $N_0 \in \nu F$ on-the-fly, building only the part of $\mathcal{G}_\varphi$ required. It works by expanding each node of $\mathcal{G}_\varphi$ until its children (line 9). Lines 6–8 take a strongly connected component (SCC) C of the expansion whose children outside C are all either in *Acc* or *Rej*, apply F inside C, accept what survives and reject the rest (Line 8). *Acc* is passed to F at line 7, so a node of C whose witness leaves C through an accepted child finds it continued there. The loop exits once N_0 is decided, and φ is satisfiable exactly when N_0 ends in *Acc*. Lines 3 and 4 are optimizations which allow pruning: a $\oplus$-node with an accepted child, or an $\otimes$-node with a rejected one, is decided on the spot, so only the components of $E \setminus (Acc \cup Rej)$ are ever computed.

Correctness of the procedure rests on three further facts: (i) pruning preserves νF; (ii) a time-progressing cycle is found by inspecting resets and bounds, making **(D3)** computable; and (iii) F may be applied inside a single component, preserving the greatest fixpoint and allowing for an on-the-fly decision.

Lemma 3. *$A \subseteq \mathcal{M}^*$ be a successful marking with $\mathcal{M}^* \cap (ch(C) \setminus C) \subseteq A$.*

(i) *If v is a $\oplus$-node with a child $w \in \mathcal{M}^*$, or an $\otimes$-node with a child $w \notin \mathcal{M}^*$, then $v \in \mathcal{M}^*$ iff $w \in \mathcal{M}^*$. Removing v's other children leaves $\mathcal{M}^*$ unchanged.*
(ii) *T contains a time-progressing cycle iff some node of T is a* (Delay)*-successor and every clock is unbounded or reset in T.*
(iii) *$\mathcal{M}^* \cap C$ is the largest $X \subseteq C$ with $X \subseteq F(X \cup A)$.*

Proof. (i) For an $\otimes$-node this is **(D2)**. For a $\oplus$-node, $\mathcal{M}^* \cup \{v\}$ is a post-fixpoint of F: v is not closed and spared by **(D2)** because of w; and **(D3)** hold at v by prefixing the edge $v \to w$ to the witnesses w carries, a pursuit active at v being either still active at w or discharged by that edge. Hence $v \in \nu F = \mathcal{M}^*$. The same prefixing gives the second claim: a path using another child of v arrives through v, so a node relying on it can take its own path to v and then v's witness. (ii) ($\Leftarrow$) Strong connectivity gives a cycle through the (Delay)-successor and a reset node of each clock T resets, which is then time-progressing. ($\Rightarrow$) By definition a time-progressing cycle elapses positive time, so it crosses a (Delay), since it is the only rule that elapses time, and every clock is either reset on it or extrapolated above M, hence unbounded. (iii) For $\subseteq$, C being maximal a witness path of $\mathcal{M}^*$ from $N \in C$ either stays in C or leaves once through some $c \in A$, where it continues since A is a successful marking. For $\supseteq$, $X \cup A$ is then a successful marking too, its A-part by monotonicity and its X-part by hypothesis, so $X \subseteq \mathcal{M}^*$ by Proposition 2.

Theorem 1. *Algorithm 1 terminates in $2^{\mathrm{poly}(n)}$ and returns* SAT *iff $N_0 \in \nu F$.*

Proof. Let $D = \mathit{Acc} \cup \mathit{Rej}$ and $U = E \setminus D$. Each iteration enlarges E or D, or the loop exits. Suppose one enlarges D by nothing; then E, D and U still carry the values they had on entry when line 9 runs. If $U = \emptyset$ then $E = D$, so $N_0 \in D$ and the loop exits. Otherwise suppose line 9 adds nothing either, that is $\mathit{ch}(U) \subseteq E$. The condensation of the subgraph induced on U is non-empty, finite and acyclic, so it has a sink C: a maximal SCC of U with $\mathit{ch}(C) \cap U \subseteq C$. Take any $w \in \mathit{ch}(C) \setminus C$. It lies outside U by sinkhood, and inside E by that supposition, hence in D. So every exit of C is decided, C meets the guard of line 5, and line 8 adds the non-empty C to D, against the assumption that D did not grow. Hence $\mathit{ch}(U) \not\subseteq E$, and line 9 enlarges E. Both sets are bounded by $|\mathcal{N}_\varphi|$, finite by Lemma 1, so at most $2|\mathcal{N}_\varphi|$ iterations enlarge either and the next one exits, necessarily with $N_0 \in D$ as that is the guard of line 10.

We show $\mathit{Acc} \subseteq \mathcal{M}^*$ and $\mathit{Rej} \cap \mathcal{M}^* = \emptyset$ by induction on the order in which nodes are accepted or rejected, each such step resting on those already taken. Lines 3 and 4 are Lemma 3(i), which also justifies leaving the remaining children unbuilt. Line 5 fires only once every child of C outside C has been accepted or rejected, so by induction $\mathit{Acc} \cap \mathit{ch}(C) = \mathcal{M}^* \cap (\mathit{ch}(C) \setminus C)$; line 7 then computes $\mathcal{M}^* \cap C$ by Lemma 3(iii), and line 8 accepts it and rejects the rest of C. Clause **(D3)** of F is evaluated through Lemma 3(ii).

For the bound let m be that of Lemma 1; as M occupies at most n input symbols, $\log m = O(n) + O(n^2) \log M = \mathrm{poly}(n)$, and each of the at most $2m$ rounds costs one decomposition of a subgraph of $\mathcal{G}_\varphi$ together with at most m applications of F, each a decomposition and $O(n)$ backward reachabilities linear in the subgraph, so the run costs $\mathrm{poly}(m) = 2^{\mathrm{poly}(n)}$.

5 Soundness and Completeness

We now prove the procedure is sound, i.e., that if the tableau succeeds on φ, then φ is SAT, and complete for non-punctual TCTL, i.e., that if φ is SAT, then the tableau succeeds on φ. These results are supported by the following lemma.

Lemma 4. *Let $N_0, \ldots, N_k$ be a path of $\mathcal{G}_\varphi$ with non-empty zones.* (i) *For every $\nu_k \in Z_k$ there are $\nu_i \in Z_i$ and delays realising every step, positive at every* (Delay) *step.* (ii) *If a delay crosses no integer hyperplane $x = b$ with $b \leq M$, every subformula of φ has constant truth on the open segment it sweeps.* (iii) *A cycle of $\mathcal{G}_\varphi$ has a non-Zeno unfolding iff it is time-progressing, and then one whose iterations elapse a constant positive amount.*

Proof. (i) Every zone operation a rule performs (intersection with a guard, reset, time-successor, $\cong_M$-canonicalisation) admits back-propagation, non-empty precisely because the successor valuation lies in its image [8]; thread backwards from ν_k, the delay of a (Delay) step being positive since $\Delta^\circ Z$ and $\Delta^\partial Z$ lie strictly ahead of Z. (ii) No clock crosses an integer at or below M, so no integer part changes and no fractional ordering wraps, while clocks above M are identified by extrapolation; the segment lies in one clock region, on which TCTL truth is constant [2,8], and its endpoints are nodes of their own by the integer split of (Delay). (iii) By contraposition, a cycle elapsing no time takes zero time over any number of traversals; and if some clock is neither reset on it nor extrapolated above M, it keeps a bound $c \leq M$, grows by exactly the delay elapsed, and caps the total at c. Conversely (i) yields a traversal elapsing some $\delta > 0$, and the cycle returning to its first node, no rule distinguishes two visits and no clock depletes a budget, so δ recurs at every iteration.

5.1 Soundness

In order to prove soundness, we proceed by constructing a witness automaton $\mathcal{A}_\varphi$ from a successful marking $\mathcal{M}^*$ of the tableau graph $\mathcal{G}_\varphi$. We then show that this automaton satisfies the formula φ.

The construction is straightforward. The saturated nodes of $\mathcal{M}^*$ supply the locations S of $\mathcal{A}_\varphi$: a saturated N carries $L(N) = \Gamma_{\mathsf{prop}}$ and $\mathsf{Inv}(N)$, and N_0 is the initial location. Every (Delay)- or (EX)-child of N that $\mathcal{M}^*$ retains supplies an outgoing edge of E, leading to the next node and carrying the reset clocks, i.e., x_μ at a (launch) and at a ($\mathbf{U}^*$-stop). Those rules elapse no time and leave L and Inv untouched, so they contribute no edge of their own.

Each location carries a schedule, which follows the finite path of $\mathcal{M}^*$ discharging one of the at most $n + 1$ obligations pending there, resolving the $\oplus$-nodes along that path and turning round-robin to the next obligation once the current one is discharged. Its memory is the pair (obligation, position on that path), so S stays finite.

Lemma 5 (Recurrence). *Let $\mu = \mathrm{A}(\varphi \,\mathbf{U}_I\, \psi)$ with $a = \inf I$, $b = \sup I < \infty$, and let ρ be a run of $\mathcal{A}_\varphi$ through a node carrying μ^*, launched at t_0. Then*

(**U****-stop) fires on ρ at times $t_0 < w_1 < w_2 < \cdots$ with $w_1 - t_0 \in I$ and $w_{j+1} - w_j \le b - a$, and $\rho \models \varphi\, \mathbf{U}_I\, \psi$ from every position at or after t_0.*

Proof. No rule drops μ^*: (**U***-stop) keeps it in Γ, (**U***-go) moves $AX\mu^*$ to $\mathcal{X}$, whence every successor carries it again, and (Abs) bars a fresh (Cov) while the pursuit is live. So μ^* rides ρ for ever, and with it the deadline on x_μ, which is b until the first witness and $b - a$ after. Both are at most M, so x_μ is bounded along ρ, and a cycle leaving a bounded clock unreset is not time-progressing. By **(M1)** the schedule reaches such a cycle and traverses it without end, and while the pursuit is live only (**U***-stop) resets x_μ, so the stop recurs: it fires at times $t_0 < w_1 < w_2 < \cdots$. At the first firing x_μ measures from t_0 and the guard is I, so $w_1 - t_0 \in I$; at each later one it measures from the preceding witness and $\mathcal{D}$ caps it at $b - a$, so $w_{j+1} - w_j \le b - a$.

For the last claim let $t \ge t_0$ and let w_j be the first witness at or after $t + a$. If $j = 1$ then $w_1 \le t_0 + b \le t + b$; otherwise $w_{j-1} < t + a$, so $w_j \le w_{j-1} + (b - a) < t + b$. Either way $w_j - t \in I$, and ψ holds at w_j while (**U***-go) asserts φ at every position before it, which is (1).

Theorem 2. *If Algorithm 1 returns* SAT *then φ is satisfiable.*

Proof. By Theorem 1 the root lies in $\mathcal{M}^*$, so $\mathcal{A}_\varphi$ is defined, and Lemma 4(i) supplies valuations and delays along its paths. It is non-blocking: its only invariants are the deadlines in $\mathcal{D}$, and cap and stop guard derive from the same interval with the same strictness, so the stop edge is enabled up to the cap and by **(M2)** the schedule takes it. By **(M1)** the schedule reaches a time-progressing cycle, which Lemma 4(iii) unfolds into a run whose iterations each elapse a fixed $\delta > 0$, so Runs^∞ is non-empty everywhere, which is all the semantics asks of it.

We show $\mathcal{T}, (s_0, \mathbf{0}) \models \varphi$ for the induced TTS by induction on φ along the marking. Propositional goals hold by saturation and closure of contradictory nodes; $EX\psi$ obtains a successor satisfying ψ from its (EX) child, and $AX\psi$ holds at all successors since each child inherits $\mathcal{X}$. For a timed modality demanded at N, (Cov) is conjunctive, so $\mathcal{M}^*$ retains both the discharge child, carrying the remaining goals, and the launch, which establishes μ with x_μ reset, so I is measured from this configuration and the pursuit rides every successor, as a goal under E and as context under A. A re-demand is covered anew, carried or absorbed, so every onset is established: by a pursuit of its own, by the carried pursuit that answers all of them, or, when the window is unbounded, by the witnesses that later launches are forced to produce. An existential until is witnessed by the path **(M2)** supplies, extended past its (**U**-stop) by a divergent run; a universal one cannot be evaded, $\mathsf{Inv}(N)$ blocking time at the cap while the pursuit is imposed on all successors, and when the demand is persistent this is Lemma 5. ($\mathbf{R}^{\mathsf{split}}$) discharges a release by partitioning the zone conjunctively: ($\mathbf{R}^\bullet$) asserts ψ in the window until φ releases it, carries the pursuit on before the window, and past it the obligation has lapsed. Lemma 4(ii) lifts each conclusion to the positions between nodes.

Theorem 3 (Completeness). *If φ is non-punctual and satisfiable then Algorithm 1 returns* SAT.

Proof. Completeness goes the other way: the realisable nodes of §3.1 are themselves a successful marking. Writing $\mathcal{R}$ for them, we show $\mathcal{R} \subseteq F(\mathcal{R})$; then $\mathcal{R} \subseteq \mathcal{M}^*$ by Proposition 2, and $N_0 \in \mathcal{R}$ as φ is satisfiable, so Theorem 1 returns SAT.

(D1): a realisable node has a non-empty zone and a consistent Γ_{prop}, and (**U**-miss) does not fire on it: a pending $\mu^\bullet$ is realisable at no valuation beyond its window, so its zone meets $x_\mu \leq b$. **(D2)**: let $N \in \mathcal{R}$ be realised by $\mathcal{T}$. All children of an $\otimes$-rule are realised by the same $\mathcal{T}$: (Cov) because $\mathcal{T}$ satisfies $\Gamma \setminus \{\mu\}$ and μ alike, ($\mathbf{R}^{\mathsf{split}}$) because its children partition Z_N and realisability is monotone under narrowing, (Delay) and (EX) because $\mathcal{T}$ admits the passage of time and supplies a successor for every $EX\psi$, each inheriting $\mathcal{X}$. At an $\oplus$-rule $\mathcal{T}$ selects a child: a disjunct it satisfies for ($\vee$); for ($\mathbf{U}^\bullet$) and ($\mathbf{U}^*$) the stop if ψ holds now with $x_\mu \in I$ and (**U**-go) otherwise, the witness being later and φ holding now; for ($\mathbf{R}^\bullet$) the stop if φ has released the obligation and the recursive child otherwise. The zone operations being exact, no chosen zone is empty. A demand re-imposed at a continuum of onsets is answered onset by onset, every re-demand being covered anew, carried or absorbed according to whether the running pursuit reaches the new onset, so the selections above serve each onset of each zone. **(D3)**: fix $N \in \mathcal{R}$ and a run ρ of $\mathcal{T}$ from a configuration for N. It is non-Zeno, so no clock stays both unreset and bounded along it, and following ρ through the choices above gives a path of $\mathcal{R}$ from N which by Lemma 1 repeats a node; that segment elapses positive time with every clock reset on it or extrapolated above M, hence is a time-progressing cycle of $\mathcal{R}$ reached from N. An until pursuit active at N is one $\mathcal{T}$ fulfils at a finite position of ρ, which the path following ρ carries it to.

Non-punctuality enters at the repetition step: a node summarises the pending obligations of a persistent universal until by x_μ alone, the time since the last witness capped at $b-a$, and by Lemma 5 that summary is faithful exactly when witnesses spaced at most $b-a$ apart answer every onset, which needs $b > a$.

Corollary 1. *Satisfiability of non-punctual TCTL is* ExpTime*-complete, and the procedure is sound and complete for TCTL of temporal nesting depth at most one, punctual intervals included.*

Proof. Membership is Theorems 1–3; hardness is inherited from CTL satisfiability [11] by fixing every interval to $[0, \infty)$. At depth one every argument is propositional, so no modality is demanded inside the pursuit of another and no fresh onset arises; the obligations live at a node are exactly the pursuits it carries, recorded in Γ, $\mathcal{X}$ and $\mathcal{D}$ with their elapsed time in Z, so a repeat agrees on the full obligation state and Theorem 3 goes through for every I.

Depth one covers reachability, safety and inevitability, the queries Uppaal [7] and Romeo [17] are asked for; the one common query beyond it, leads-to $\mathrm{AG}(p \to \mathrm{AF}_I q)$, stays covered by Theorem 3 whenever I is non-punctual.

Table 1: **RQ1: cost drivers.** One row per family and verdict, over the whole sweep. The growth law is fitted to the *node count*, which is a deterministic function of the formula; wall-clock is reported as a maximum only, since instrumentation floors it at 0.2 s. No instance timed out or reached the memory cap.

Factor	Verdict	Range	Nodes	R^2	max nodes	max t (s)	max mem (MB)
Clocks k	SAT	1–20	$\mathbf{1.93^k}$	1.00	4,194,836	494.57	3628
Clocks k	UNSAT	1–20	linear	0.99	279	0.20	2
Depth d	UNSAT	1–12	$d^{2.48}$	1.00	211,527	11.16	178
Depth d	SAT	1–12	$d^{1.84}$	0.98	10,283	0.49	9
Size m	SAT	1–1024	$m^{0.63}$	0.94	1,038	0.82	32
Const M	SAT	2–1048576	flat	—	286	0.20	4
Const M	UNSAT	1–1048576	flat	—	401	0.20	2

6 Evaluation

We evaluate the procedure through two research questions.

RQ1: Which parameters of the input formula drive the cost of the search, and how does it scale in each?

RQ2: Is the procedure practical on the flat, one-level TCTL queries engineers write for Uppaal?

RQ1 asks where the ExpTime bound of Theorem 1 is felt and where it is not; **RQ2** asks the complementary question, whether the queries a specification is actually checked against lie in the expensive regime.

Benchmark and method. No standard suite exists for TCTL *satisfiability*, and there is no second implementation to compare against, so the study characterises one procedure rather than ranking two. We generate 333 parameterised instances. For **RQ1** each family varies a single term of $2^{O(n)} \cdot M^{O(n^2)}$ with the others fixed: the number of *simultaneously live* formula-clocks k, obtained by conjoining k independent timed untils; the *nesting depth* d of a timed double-check; the largest constant M, swept by powers of two to 2^{20}; and the propositional *size* m. Each ships matched satisfiable and unsatisfiable variants, so witness cost and refutation cost are separated. For **RQ2** we generate the canonical Uppaal shapes over atomic propositions, each swept along payload size $m \leq 1024$ and time bound $b \leq 2^{20}$.

We report the number of tableau *nodes* as the primary measure. It is a deterministic function of the formula, independent of the machine, and it is what Lemma 1 bounds; every growth law below is fitted to it. Wall-clock time and peak resident memory are reported as maxima only. Each instance was given 600 s and a 10 GB resident-memory cap, run serially on an Apple M4 Pro; timing is the median of three runs for instances under 5 s and a single run above. **No instance timed out, and none reached the memory cap.**

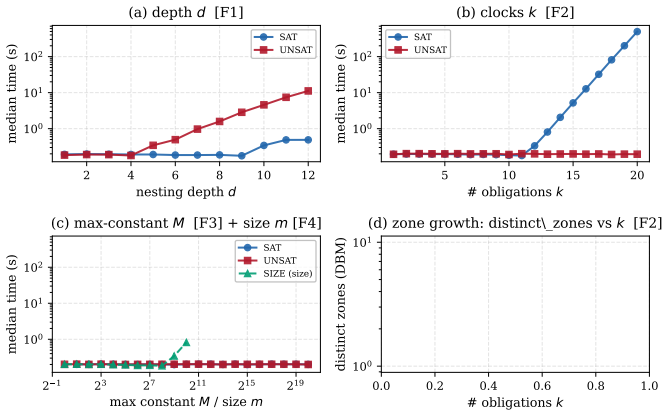


Fig. 3: **RQ1** cost drivers: median runtime against each parameter, satisfiable and unsatisfiable shown separately. Clock simultaneity is exponential, depth is polynomial, and the constant M and the propositional size are flat.

***RQ1**: one exponential driver, and it is not nesting.* Table 1 ranks the four factors, and they separate sharply.

Clock simultaneity is the only exponential axis. On satisfiable instances the node count is 1.93^k with $R^2 = 1.00$ over the whole sweep $k = 1..20$, a clean doubling per added clock, ending at 4.19M nodes, 3.6 GB and 495 s. This is the $M^{O(n^2)}$ handle: k clocks live at once means a DBM of dimension $k + 1$, and a witness has to meet all k windows together, so the search enumerates their product.

Nesting depth is polynomial. Refutation grows as $d^{2.48}$ ($R^2 = 1.00$) and a witness as $d^{1.84}$, so depth 12 costs 211k nodes and 11 s unsatisfiable, 10k nodes and half a second satisfiable. Nesting is an order of magnitude cheaper than the bound permits, because the memoised graph collapses the repeated structure of the nested check instead of unfolding it.

Neither M nor formula size is a driver. The node count is constant at 293 across six orders of magnitude of M, from 1 to 2^{20}: a zone represents a large bound symbolically, and $\cong_M$ leaves the realised set identical. Propositional size is sublinear, $m^{0.63}$, so a payload of 1024 atoms realises 1038 nodes. The closure grows; the search does not.

The witness/refutation asymmetry runs in both directions, and which way depends on what the formula demands. On clocks the satisfiable side is the expensive one, 4.19M nodes against 279: a model must meet every obligation at once, whereas a single contradiction closes the refutation outright. On depth the order reverses, 211k against 10k: the witness is short-circuited at the first accepting branch, while the refutation has to exhaust the graph. The ranking is therefore *clocks* $\gg$ *depth* $\gg \{M, \mathit{size}\}$, with the exponential confined to satisfiable instances of the first.

Table 2: **RQ2: the Uppaal query shapes.** Worst case over the full sweep of payload size $m \in [1, 1024]$ and time bound b, per shape. All 217 instances were solved; none timed out.

Shape	TCTL form	b range	#	max nodes	max t (s)	max mem (MB)
Reachability	$\mathrm{EF}_{[0,b]}\,\Phi$	1–2^{20}	52	1,030	0.50	27
Safety	$\mathrm{AG}_{[0,b]}\,\Phi$	1–2^{20}	52	3,091	0.51	28
Bounded liveness	$\mathrm{AF}_{[1,b]}\,\Phi$	2–2^{20}	51	1,038	0.51	27
Leads-to	$\mathrm{AG}(p \rightarrow \mathrm{AF}_{[1,b]}q)$	2–2^{20}	51	15,489	1.14	39
Possibly-always	$\mathrm{EG}_{[0,\infty)}\,\Phi$	—	11	2,059	0.82	33

***RQ2**: the queries engineers write are far from the wall.* All 217 instances of the Uppaal sweep are solved, none times out, and the worst case over every shape and both axes is 1.14 s, 15,489 nodes and 39 MB (Table 2). The time bound is absorbed entirely: swept to $b = 2^{20}$, the hardest shape realises 1242 nodes and stays at the 0.2 s floor, which is the constant-independence of **RQ1** showing up again. Only the payload size moves anything, and mildly. Leads-to, $\mathrm{AG}(p \rightarrow \mathrm{AF}_{[1,b]}q)$, is the costliest shape by an order of magnitude in nodes, as its nesting depth of two predicts, and is still solved in about a second. Why these shapes are cheap is visible in **RQ1**: each keeps one formula-clock live, so the one exponential axis is never entered.

Threats to validity. The benchmark is synthetic; the Uppaal family mitigates this over the canonical query *shapes*, but on abstract propositions rather than ones drawn from deployed models. No wall was located: with no timeouts, the largest costs reported are the ends of the sweeps rather than limits of the procedure, and the 1.93^k law puts $k \approx 24$ past the memory cap. Wall-clock below 0.2 s is dominated by process startup and by the 150 ms sampling interval of the memory watchdog, which is why the fits use node counts; that watchdog also makes the cap a sampled bound rather than a hard one, macOS implementing no address-space limit. Timings come from one machine and one zone backend, and the generator, solver and a reproducibility manifest accompany the paper.

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